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Date of Experiment: 17/10/2024 & 24/10/2024

CM3191 Chemical Synthesis Experiments

Organic Experiment 2: Functionalized 1,2-diphenylethyne Synthesis by Sonogashira Coupling and Purification by Column Chromatography

1. Aim

To synthesize a functionalized 1,2-diphenylethyne via Sonogashira coupling of aryl iodide and ethynylbenzene using a Pd/C catalyst. To identify the suitable solvent ratio based on R_f value differences determined by thin layer chromatography (TLC) and perform column chromatography to isolate the pure product. Finally, to confirm the product purity and the identity of its functional group based on ^{1}H NMR analysis.

2. Results and Discussion

2.1 TLC

2.1.1 Reaction time determination

Using the solvent ratio of 5% ethyl acetate (EA) and 95% hexane, the TLC was performed 1 hour after setting the reaction. The point for ethynylbenzene was clearly visible in the reaction mixture (R1). Hence, the TLC was performed again 30 minutes later and ethynylbenzene was no longer clearly visible in the reaction mixture (R2)

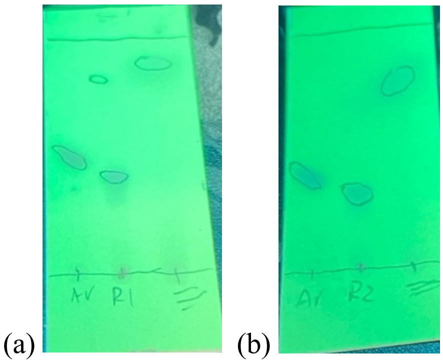

Figure 1: Image of the TLC plates (5% EA and 95% hexane) under UV light after heating for (a) 1 hour and (b) 1.5 hours (Starting from the left, each point is aryl iodide, reaction mixture and ethynylbenzene)

However, it was not evident whether the point of R2 is the product or the aryl iodide (Figure 1(b)). The point's distance from baseline was similar to that of aryl iodide. Therefore, separation for a crosspoint was tested out by spotting both the aryl iodide and R2 at the same point on the baseline. Because two points were observed at the crosspoint, it was confirmed that the reaction mixture mainly consists of the product instead of the aryl iodide. Noting that aryl iodide is mostly consumed, the reaction was stopped after heating for 1.5 hours.

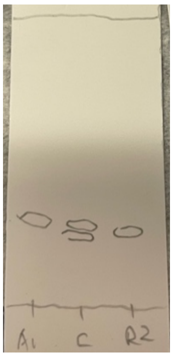

Figure 2: Image of the TLC plate (5% EA and 95% hexane)
(Starting from the left, each point is aryl iodide, crosspoint and reaction mixture after 1.5 hours)

2.1.2 R_f value calculation

	1% EA	5% EA
Ethynylbenzene	0.5428	0.6857
Functionalized 1,2-diphenylethyne (product)	0.1714	0.3143
Aryl iodide	0.1143	0.2571

Table 1: R_f values for 1% and 5% EA solvent ratios

To determine the suitable solvent ratio for column chromatography, R_f values were calculated for different solvent ratios. For both 1% and 5% EA, ethynylbenzene was clearly separated from the product with the R_f value difference being 0.3714. The R_f value difference between the product and the aryl iodide was also identical for both solvent ratios, 0.0571 and 0.0572 for 1% and 5% EA, respectively. These calculations suggest that both solvent ratios would result in the same efficiency in separation. However, 5% EA was chosen for column chromatography due to the effect of mobile phase polarity on retention time.

Retention time will be extended with lower percentage of EA, the polar solvent, because the polar compounds will adsorb to the silica gel more strongly and for a longer duration. Thus, mobile phase with too low polarity cannot effectively separate and elute out polar compounds. This is undesirable since longer duration of being adsorbed to the silica gel results in a longer retention time and greater longitudinal diffusion. The bands for the product and the aryl iodide will merge because of diffusion and further increase the separation difficulty which is already challenged by having similar R_f values. Based on these factors, the literature recommends choosing a solvent ratio that gives $R_f = 0.35$ for the compound that is desired to be separated[1]. For 5% EA, the R_f value of the product ($R_f = 0.3143$) is significantly closer to 0.35 compared to that of 1% EA. As a result, 5% EA and 95% hexane was chosen for column chromatography.

Simply observing the TLC results and comparing the distance travelled, the resolution of the product and aryl iodide was not evident. The product should have a higher R_f value than aryl iodide considering two factors: less polar compounds can travel farther from the baseline in 95% hexane and the product has a lower polarity than the aryl iodide due to the additional aromatic ring. Nonetheless, both Figures 1 and 2 indicate that aryl iodide can travel farther from the baseline than the product. Such observation is due to the edge effect where solvent front travels unevenly closer to the TLC plate edge and causes point distortion[2]. To prevent this edge effect, following practices can be done. Firstly, developing chamber that fits the TLC plate size can be used to prevent the plate from tilting and causing solvent to travel up more quickly near the edges. Such choice will make it easier to position the TLC plate so that the baseline is parallel to the solvent level. Additionally, samples can be spotted closer to the middle of the baseline.

2.2 Column chromatography

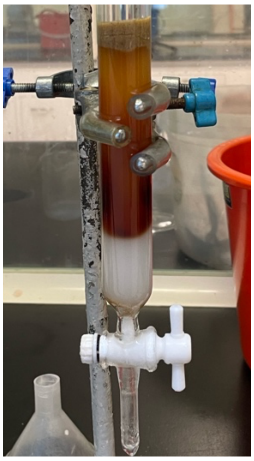

Figure 3: Separation of different coloured bands during column chromatography (5% EA and 95% hexane)

Based on the TLC results, column chromatography was performed using 5% EA and 95% hexane. Two distinct bands were observed as shown in Figure 3 where the first dark brown band to be eluted out belong to ethynylbenzene and the second lighter band is a mixture of the product and aryl iodide. Figure 4 shows that test tube 1 contains all three compounds, test tubes 2 to 4 contains both the product and aryl iodide, and test tube 5 only contains aryl iodide.

Such observations once again confirmed the idea that aryl iodide traveling farther from the baseline (Figure 1) was simply due to the edge effect. The higher lying point is the major component which must be the product noting that Sonogashira coupling is a high yielding reaction (further discussion in Section 2.4). Moreover, Figure 5 shows that brown liquid resulted after concentrating the test tube 5 eluent in vacuo. Since the product must be solid, it suggests that aryl iodide is indeed a minor component with a smaller R_f value.

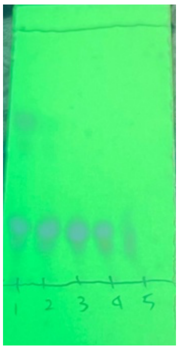

Figure 4: Image of the TLC results for column chromatography (5% EA and 95% hexane)

Figure 5: Image of the round bottom flask containing aryl iodide concentrated from test tube 5

Column chromatography was performed again after collecting test tubes 2 to 4 and concentrating in vacuo. The solvent ratio of 1% EA was used for the second attempt to test whether product can be better separated. However, the separation was unsuccessful and all the test tubes contained both the product and the aryl iodide. Therefore, all eluent was collected and submitted for ^1H NMR analysis.

Lowering the solvent polarity was insufficient to obtain the pure product. Resolution can be improved by increasing the number of theoretical plates, the measure of column efficiency and the theoretical unit at which equilibrium partitioning of compounds occurs between mobile and stationary phases[3]. The two simple methods of increasing the number of theoretical plates are packing the column height to be longer and using silica gel of smaller diameter. Increasing the column length leads to a greater overall surface area of the stationary phase where compounds can adsorb. Hence, more partition equilibria are established, and the total number of theoretical plates is increased. Reducing the silica gel diameter has a similar effect of increasing the surface area of the stationary phase per particle. Resolution is enhanced as polar compounds more readily adsorb to the stationary phase while less polar compounds are retained in the mobile phase to be eluted out. Therefore, packing the column longer than 10 cm and using silica gel smaller than 50 micron would enable easier separation of the product. However, it must be noted that both approaches are accompanied with undesired effects such as greater longitudinal diffusion. Adequate parameters must be tested to confirm the combination resulting in the greatest resolution rather than ambiguously increasing the column length or reducing silica gel diameter.

2.3 ^1H NMR analysis

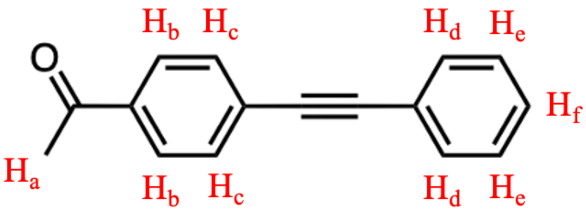

Figure 6: Chemical structure of 1-[4-(phenylethynyl)phenyl]ethenone with the assignment of protons

Chemical shift (ppm)	Multiplicity	Coupling constant (Hz)	Integration	Assignment
7.96-7.93	m	-	2.07	H_b
7.63-7.60	m	-	2.07	H_c
7.57-7.54	m	-	2.04	H_d
7.38-7.36	m	-	3.19	H_e & H_f
2.61	s	-	3.00	H_a

Table 2: ^{1}H NMR peak assignment for protons of 1-[4-(phenylethynyl)phenyl]ethenone

Five peaks were identified to be belonging to the functionalized 1,2-diphenylethyne. The peaks clustered between 7.96-7.36 ppm correspond to aromatic protons which do not provide detailed information about the functional group. Instead, the critical peak is observed at 2.61 ppm. Amongst the four possible functional groups - H, NO_2, COMe and COOMe - it was first identified that H and NO_2 could not be the functional groups. If it were H, no additional peak would have appeared since the proton would be chemically equivalent to H_f shown in Figure 6. Similarly, NO_2 does not introduce new protons which contradict to the sharp singlet peak appearing at 2.61 ppm. If COOMe were the functional group, the singlet peak must be around 4 ppm because protons of the methoxy group are more de-shielded due to the inductive effect of the ester compared to a ketone with only a carbonyl group[4]. As a result, COMe was confirmed to be the functional group which aligns with the literature reporting the H_a peak at 2.63 ppm[5]. The four set of aromatic proton peaks were assigned based on the fact that chemical shift value decreases as the proton is farther away from the polar ketone group that causes de-shielding. Thus, the starting material was identified to be 4-iodoacetophenone and the product would be 1-[4-(phenylethynyl)phenyl]ethenone.

Several peaks were observed in addition to those belonging to 1-[4-(phenylethynyl)phenyl]ethenone, which were identified to be solvent peaks instead of an impurity. A singlet peak at 5.29 ppm indicated CH_2Cl_2. A doublet at 1.25 ppm, a singlet at 2.04 ppm and a quartet at 4.12 ppm are all attributed to EA. In addition, a short and broad singlet peak at 1.61 ppm corresponds to water[6]. A minor peak at 2.17 ppm indicates the presence of acetone, likely have been introduced when rinsing the round bottom flask used to concentrate the test tube 5 eluent (Section 2.2). These peaks could have been removed by prolonged drying under vacuum.

The absence of peak around 3 ppm indicates that ethynylbenzene is fully removed[5]. Furthermore, the peak intensity values closely matches that of the number of protons of 1-[4-(phenylethynyl)phenyl]ethenone. If there were significant amount of aryl iodide left unseparated, the integration value for the peak at 7.96-7.93 ppm would have been significantly greater than 2.07. Moreover, an additional multiplet peak may have been observed for protons at the ortho position of iodine. Nevertheless, it cannot be concluded that the purification by column chromatography was successful. TLC results show that both 1-[4-(phenylethynyl)phenyl]ethenone and 4-iodoacetophenone are present in the final product after column chromatography. The presence may have been masked in the ^{1}H NMR spectrum because Sonogashira coupling is a high yielding reaction, leaving only trace amount of starting material after reaction completion. If the yield was low and large amounts of starting materials remained, the ^{1}H NMR spectrum would have shown the peaks for aryl iodide more prominently and suggest unsuccessful separation. Thus, as discussed in Section 2.2, the parameters for column chromatography can be adjusted to improve the separation efficiency.

2.4 Percentage yield

The ^{1}H NMR analysis confirmed that the functional group is COMe. Hence, the percentage yield of purified 1-[4-(phenylethynyl)phenyl]ethenone was calculated to be 79.81% (detailed calculation in Section 5.2). The yield is significantly lower than the literature yield of 95%[5]. Such low yield can be attributed to two factors. Due to the small reaction scale, the reaction was performed in a small vial with only 1 mL of GVL solvent. As shown in Figure 7, this resulted in a fair portion of the reaction mixture to adhere to the vial's side wall when stirred at high speed. This portion of the reaction mixture could not react to form 1-[4-(phenylethynyl)phenyl]ethenone, contributing to the lower yield. Moreover, a notable difference was that anhydrous solvent was employed and the solvent was purged with nitrogen gas in the literature. Since the Pd/C catalyst is sensitive to both oxygen and moisture[7], the use of anhydrous and deoxygenated GVL solvent would improve catalyst's stability and activity, resulting in a greater yield for the same reaction duration.

Figure 7: Image of the vial after reaction completion

3. Conclusion

Sonogashira coupling was carried out successfully to obtain functionalized 1,2-diphenylethyne which was identified to be 1-[4-(phenylethynyl)phenyl]ethenone according to the ^{1}H NMR analysis. Though the ^{1}H NMR spectrum depicted that aryl iodide was not present, the TLC results indicated that aryl iodide did exist as a minor component in the final product obtained after column chromatography. Since simply adjusting the solvent ratio was insufficient to separate the two compounds, increasing the column length and using silica gel of smaller diameter can be considered for subsequent column chromatography. The percentage yield of 79.81% is lower than the literature results and can be improved by ensuring that all reactants are immersed in anhydrous and deoxygenated GVL solvent.

4. References

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5. Appendix

5.1 ^{1}H NMR

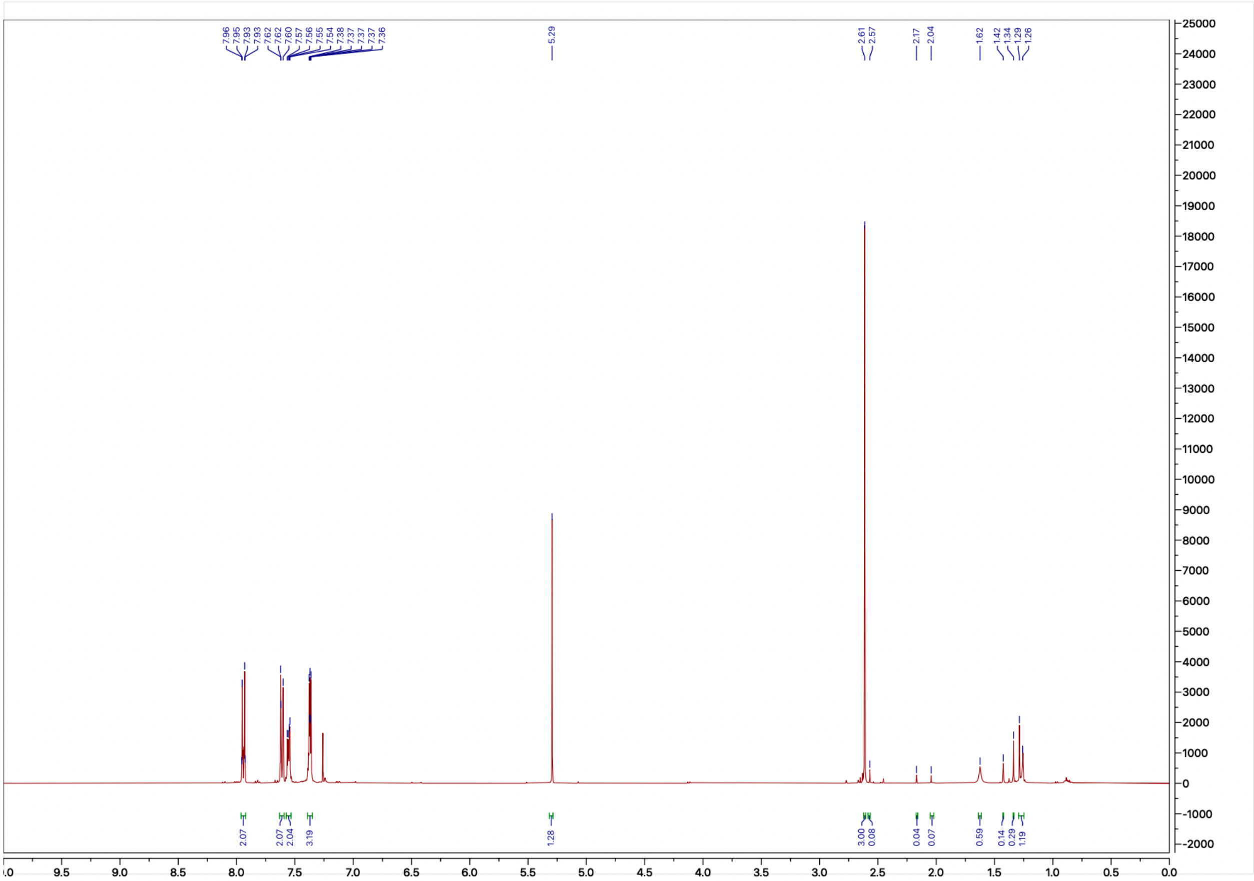
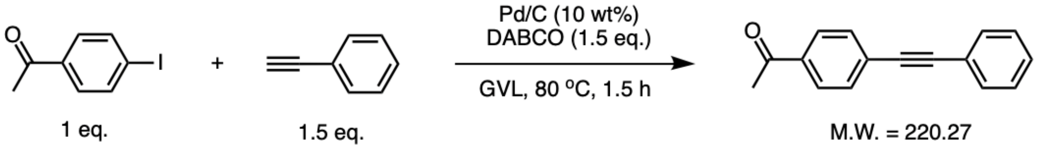


Figure A1: ^{1}H NMR spectrum of 1-[4-(phenylethynyl)phenyl]ethenone (400 MHz, $CDCl_3$, 298 K)

5.2 Percentage yield calculation

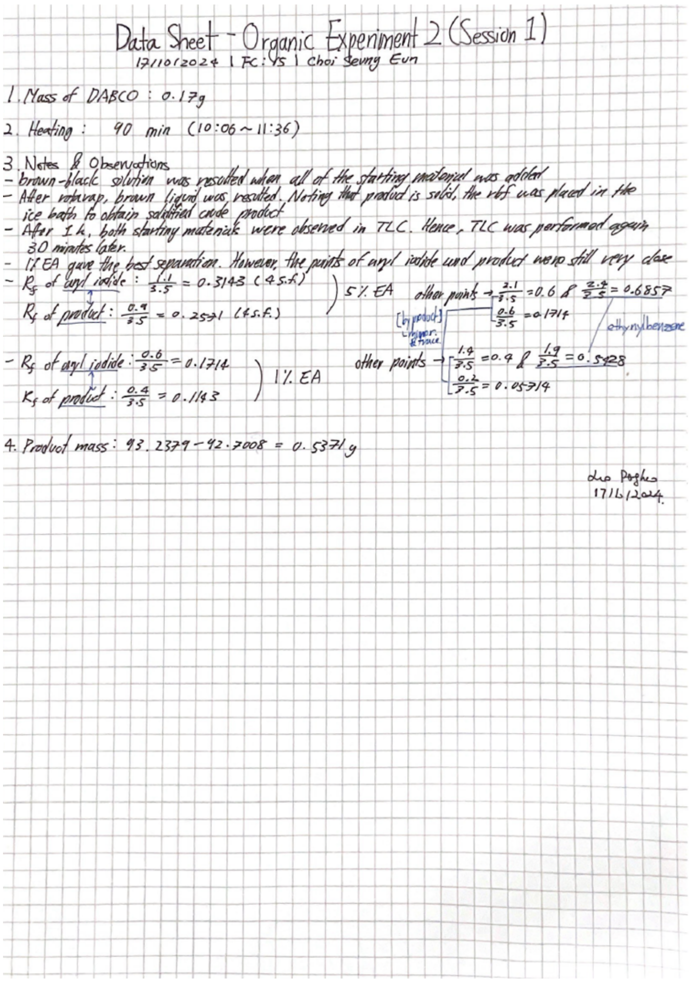


Scheme A1: Sonogashira coupling reaction with chemical structures and equivalence shown

Mass of the crude product:
93.2379 (g) – 92.7008 (g) = 0.5371 (g)
Moles of the crude product:
0.5371 (g) ÷ 220.27 (g/mol) = 2.438 mmol (4 s.f.)
Percentage yield of the crude product:
2.438 (mmol) ÷ 1.000 (mmol) × 100% = 243.8% (4 s.f.)

Mass of the purified 1-(4-(phenylethynyl)phenyl)ethenone:
85.9797 (g) – 85.8039 (g) = 0.1758 (g)
Moles of the purified 1-(4-(phenylethynyl)phenyl)ethenone:
0.1758 (g) ÷ 220.27 (g/mol) = 0.7981 mmol (4 s.f.)
Percentage yield of 1-(4-(phenylethynyl)phenyl)ethenone:
0.7981 (mmol) ÷ 1.000 (mmol) × 100% = 79.81% (4 s.f.)

5.3 Data sheets



Data Sheet - Organic Experiment 2 (Session 1)
17/10/2024 | FC: YS | Choi Seung Eun

1. Mass of DABCO : 0.17g

2. Heating : 90 min (10:06 ~ 11:36)

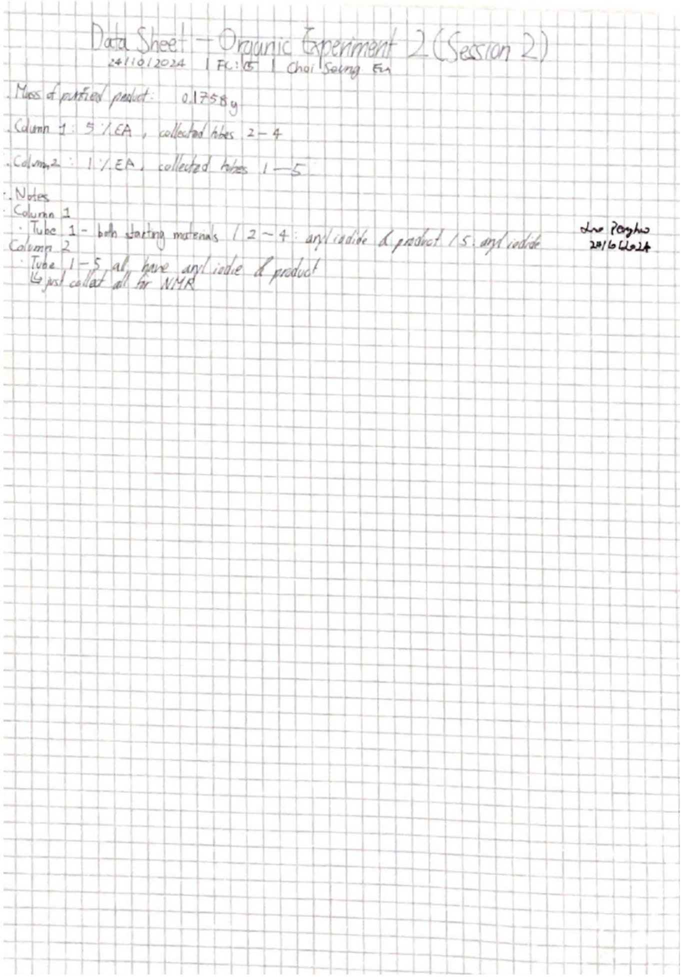
3. Notes & Observations
- brown-black solution was resulted when all of the starting material was added
- After rotavap, brown liquid was resulted. Noting that product is solid, the rbf was placed in the ice bath to obtain solidified crude product
- After 1 h, both starting materials were observed in TLC. Hence, TLC was performed again 30 minutes later.
- 1% EA gave the best separation. However, the points of aryl iodide and product were still very close
- R_f of aryl iodide : $\frac{1.1}{3.5} = 0.3143$ (4 s.f.)
  R_f of product : $\frac{0.9}{3.5} = 0.2571$ (4 s.f.)
) 5% EA other points → $\frac{2.1}{3.5} = 0.6$ & $\frac{2.4}{3.5} = 0.6857$
  $\frac{0.6}{3.5} = 0.1714$
- R_f of aryl iodide : $\frac{0.6}{3.5} = 0.1714$
  R_f of product : $\frac{0.4}{3.5} = 0.1143$
) 1% EA other points → $\frac{1.4}{3.5} = 0.4$ & $\frac{1.9}{3.5} = 0.5428$
  $\frac{0.2}{3.5} = 0.05714$

[by product] (minor & trace)
ethynylbenzene

4. Product mass : 93.2379 - 92.7008 = 0.5371 g

17/10/2024.

Figure A2: Data sheet of session 1 (17/10/2024)



Data Sheet – Organic Experiment 2 (Session 2)
24/10/2024 | FC: 05 | Choi Seung Eun

Mass of purified product: 0.1758 g

Column 1: 5 % EA, collected tubes 2 – 4

Column 2: 1 % EA, collected tubes 1 — 5

Notes
- Column 1
  - Tube 1 - both starting materials / 2 ~ 4: aryl iodide & product / 5: aryl iodide
- Column 2
  - Tube 1 – 5 all have aryl iodie & product
    ↳ just collect all for NMR

24/10/2024

Figure A3: Data sheet of session 2 (24/10/2024)